



Global natural wetland methane emissions (2000–2025)

Mengze Li^{1,2}, Robert B. Jackson^{2,3}, Marielle Saunois⁴, Philippe Ciais⁴, Ben Poulter⁵, Josep G. Canadell⁶,
 Prabir K. Patra^{7,8}, Hanqin Tian^{9,10}, Zhen Zhang¹¹, Etienne Fluets-Chouinard¹², Zutao Ouyang¹³, Ting
 5 Zhang¹, David J. Beerling¹⁴, Dmitry A. Belikov¹⁵, Philippe Bousquet⁴, Danilo Custodio⁴, Naveen
 Chandra⁷, Xinyu Dou², Nicola Gedney¹⁶, Peter O. Hopcroft¹⁷, Alison M. Hoyt², Kazuhito Ichii^{15,18},
 Akihito Ito¹⁹, Atul K. Jain²⁰, Katherine Jensen²¹, Fortunat Joos²², Thomas Kleinen²³, Masayuki
 Kondo^{8,24}, Fa Li², Tingting Li²⁵, Xiangyu Liu²⁶, Shamil Maksyutov²⁷, Avni Malhotra²⁸, Adrien
 10 Martinez⁴, Kyle McDonald²¹, Joe R. Melton²⁹, Paul Miller³⁰, Jurek Müller²², Yosuke Niwa^{27,31}, Shufen
 Pan⁹, Shushi Peng³², Changhui Peng^{33,34}, Zhangcai Qin³⁵, Peter Raymond³⁶, William Riley³⁷, Arjo
 Segers³⁸, Rona L. Thompson¹⁶, Aki Tsuruta³⁹, Xi Yi⁴, Kunxiaoia Yuan⁴⁰, Wenxin Zhang³⁰, Bo
 Zheng^{41,42}, Qing Zhu³⁷, Qian Zhu⁴³, Qianlai Zhuang²⁶

- 1 Department of Geography, National University of Singapore, Singapore.
- 15 2 Stanford Doerr School of Sustainability, Department of Earth System Science, Stanford, CA, USA.
- 3 Department of Earth System Science, Woods Institute for the Environment, and Precourt Institute for Energy, Stanford University, Stanford, CA, USA.
- 4 Laboratoire des Sciences du Climat et de l'Environnement (LSCE), CEA, CNRS, UVSQ, Université Paris Saclay, Gif-sur-Yvette, France.
- 20 5 Spark Climate Solutions, San Francisco, CA, USA.
- 6 Global Carbon Project, CSIRO Environment, ACT 2601, Australia.
- 7 Research Institute for Global Change, JAMSTEC, 3173-25 Showa-machi, Kanazawa, Yokohama, 236-0001, Japan.
- 8 Seto Inland Sea Carbon Neutral Research Center (S-CNC), Hiroshima University, Higashi-Hiroshima, Hiroshima 739-8529, Japan.
- 25 9 Center for Earth System Science and Global Sustainability, Schiller Institute for Integrated Science and Society, Boston College, Chestnut Hill, MA, USA.
- 10 Department of Earth and Environmental Sciences, Boston College, Chestnut Hill, USA.
- 11 State Key Laboratory of Tibetan Plateau Earth System, Environment and Resources, and Institute of Tibetan Plateau Research, Chinese Academy of Sciences, Beijing, China.
- 30 12 Earth System Science Division, Pacific Northwest National Laboratory, Richland, WA, USA.
- 13 College of Forestry, Wildlife and Environment, Auburn University, Auburn, AL, USA.
- 14 School of Biosciences, University of Sheffield, U.K.
- 15 Center for Environmental Remote Sensing, Chiba University, Chiba, Japan.
- 16 MetOffice Hadley Centre, Joint Centre for Hydrometeorological Research, Wallingford, U.K.
- 35 17 School of Geography, Earth & Environmental Sciences, University of Birmingham, U.K.
- 18 Graduate School of Science and Engineering, Chiba University, 1-33, Yayoi-cho, Inage-ku, Japan.
- 19 Graduate School of Agricultural and Life Sciences, The University of Tokyo, Tokyo, Japan.
- 20 Department of Climate, Meteorology, and Atmospheric Sciences, University of Illinois at Urbana-Champaign, Urbana, IL, USA.
- 40 21 Department of Earth and Atmospheric Sciences, City University of New York, New York, NY, USA.
- 22 Climate and Environmental Physics, Physics Institute and Oeschger Centre for Climate Change Research, University of Bern, Bern, Switzerland.
- 23 Max Planck Institute for Meteorology, Hamburg, Germany.
- 24 The IDEC Institute - Center for Peaceful and Sustainable Futures (CEPEAS), Network for Education and Research on
- 45 Peace and Sustainability (NERPS), Graduate School of Humanities and Social Sciences International Economic



Development Program (IEDP), Graduate School of Innovation and Practice for Smart Society (SmaSo), Hiroshima University, Higashi-Hiroshima, Hiroshima 739-8529, Japan.

25 Key Laboratory of Atmospheric Environment and Extreme Meteorology, Institute of Atmospheric Physics, Chinese Academy of Sciences, Beijing, China.

50 26 Department of Earth, Atmospheric, and Planetary Sciences, Department of Agronomy, Purdue University, West Lafayette, IN, USA.

27 Center for Global Environmental Research, National Institute for Environmental Studies, Tsukuba, Ibaraki, Japan.

28 Biological Sciences Division, Pacific Northwest National Laboratory, Richland, WA, USA.

29 Climate Research Division, Environment and Climate Change Canada, Victoria, BC, Canada.

55 30 Department of Physical Geography and Ecosystem Science, Lund University, Sölvegatan 12, 223 62, Lund, Sweden.

31 Meteorological Research Institute (MRI), Nagamine 1-1, Tsukuba, Ibaraki 305-0052, Japan.

32 Sino-French Institute for Earth System Science, College of Urban and Environmental Sciences, Peking University, Beijing 100871, China

60 33 Department of Biology Sciences, University of Quebec at Montreal, C.P. 8888, Succ. Centre-Ville, Montreal, QC H3C 3P8, Canada.

34 College of Geographic Science, Hunan Normal University, Changsha 410081, China.

35 School of Atmospheric Sciences, Sun Yat-sen University, and Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai), Zhuhai 519000, China.

36 School of the Environment, Yale University, New Haven, CT, USA.

65 37 Climate and Ecosystem Sciences Division, Lawrence Berkeley National Laboratory, Berkeley, CA, USA.

38 TNO, Department of Climate Air & Sustainability, P.O. Box 80015, NL-3508-TA, Utrecht, the Netherlands.

39 Finnish Meteorological Institute, P.O. Box 503, 00101, Helsinki, Finland.

40 Department of Earth and Atmospheric Sciences, University of Houston, Houston, TX, USA.

70 41 Institute of Environment and Ecology, Tsinghua Shenzhen International Graduate School, Tsinghua University, Shenzhen 518055, China.

42 State Environmental Protection Key Laboratory of Sources and Control of Air Pollution Complex, Beijing 100084, China.

43 College of Hydrology and Water Resources, Hohai University, Nanjing 210098, China.

Correspondence to: Mengze Li (mengze@nus.edu.sg)

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80 **Abstract.** Wetlands are the largest natural source of atmospheric methane (CH₄), yet comprehensive
 global budgets are typically delayed by several years, preventing a timely understanding of CH₄ sources,
 sinks, and their trends. To reduce this delay, we present a model emulator-driven framework and
 accompanying workflow that enable timely, continuous emission updates and applying the framework
 to a global dataset of natural vegetated wetland CH₄ emissions to extend the most recent Global
 85 Methane Budget (GMB; Saunio et al., 2025) record through 2025 at monthly 1°x1° resolution. We
 developed a machine-learning emulator to reconstruct spatially explicit monthly emission fields (global
 $R^2 = 0.65 \pm 0.003$ (mean $\pm$ 95% CI, hereafter) and $RMSE = 5.49 \pm 0.12 \times 10^{-3}$ Tg CH₄/year in test data
 which is ~30% of the total data). The emulator is trained on 35 GMB model estimates (22 process-based
 model estimates and 13 atmospheric inversion estimates) paired with 10 ensemble realizations of 11
 90 gridded climate predictor variables from atmospheric reanalyses. While the global mean predicted
 wetland CH₄ emissions for 2021-2025 (157.83 ± 2.38 Tg CH₄/year) are only marginally higher (~0.05
 Tg CH₄/year) than the 2000-2020 baseline, this stability masks a significant hemispheric redistribution



of emissions. We detect a surge in Northern Hemisphere emissions in 2021-2025, with mid- and high-latitudes increasing by 0.76 ± 0.07 (z-score: 2.21) and 0.35 ± 0.03 Tg/year (z-score: 1.01), respectively, while the tropics and Southern Hemisphere extratropics show offsetting negative trends (-0.95 ± 0.19 and -0.11 ± 0.02 Tg/year with z-scores of -2.81 and -0.34, respectively). The predicted emissions capture the low emissions in 2023 in South America linked to El Niño-related drought, as reported by recent studies (Ciais et al., 2026; Quinn et al., 2025). Post-2020 growth rates of emission anomalies are a magnitude higher than that in 2000-2025, suggesting an intensification of emission variability. Furthermore, we identify a distinct seasonal amplification of global emission growth peaking in late boreal summer. This new dataset and operational framework bridge the gap between latest updated budgets and low-latency monitoring, providing a scalable capacity to frequently update global emission estimates and critical early warnings of regional wetland feedback loops. The data are publicly available at <https://doi.org/10.5281/zenodo.18870108> (Li et al., 2026).

1 Introduction

Wetlands are the largest natural source of methane (CH_4), a potent greenhouse gas. They account for 25-30% of global CH_4 emissions and strongly influence the global carbon cycle (Jackson et al., 2024; Peng et al., 2022; Saunois et al., 2025; Zhang et al., 2023). Quantifying how wetland CH_4 emissions respond to climate variability is essential for interpreting interannual variability in the global CH_4 budget and for understanding recent atmospheric CH_4 changes (Ciais et al., 2026; Gedney et al., 2019a; Parker et al., 2018; Stavert et al., 2022). Community efforts such as the Global Methane Budget (GMB) synthesize process-based “bottom-up” (BU) models and atmospheric “top-down” (TD) inversions to estimate all CH_4 emission sources and sinks (Saunois et al., 2020, 2025). However, the significant effort required to assemble these products into a trustworthy ensemble typically leads to lags of multiple years. For instance, the most recent GMB published in 2025 (Saunois et al., 2025) covers the period 2000-2020. Although low-latency estimates exist for some individual models (Quinn et al., 2025), accelerating the assembly of a trustworthy multi-model synthesis would better support timely emission monitoring and evidence-based assessment of mitigation progress.

Wetland CH_4 emissions are governed by coupled biogeochemical and hydrometeorological controls that regulate CH_4 production and oxidation in the sediment and water column and transport to the atmosphere. Temperature exerts a strong influence on microbial methanogenesis and ecosystem respiration (Bansal et al., 2023; Li et al., 2025; Yvon-Durocher et al., 2014), while soil state can modulate growing-season length and cold-season dynamics (Hyvärinen et al., 2025; Zona et al., 2016). Hydrology, often summarized by water table depth, soil moisture and inundation, controls oxygen availability and redox conditions, thereby influencing wetland production and oxidation (Cui et al., 2024; He et al., 2025; Knox et al., 2021). Precipitation and surface energy fluxes jointly constrain water balance through inputs and evapotranspiration (Aalto et al., 2025; Helbig et al., 2020; Tyystjärvi et al., 2024), while radiation and vegetation state provide proxies for substrate supply and plant-mediated



transport pathways that connect below-ground CH₄ production to the atmosphere (Helfter et al., 2022; McNicol et al., 2023).

- Currently, global wetland CH₄ emissions are primarily estimated using two modeling approaches:
- 135 bottom-up (BU) models and top-down (TD) models. Process-based BU models explicitly simulate the complex biogeochemical mechanisms driven by environmental conditions (Zhang et al., 2025). TD atmospheric inversions estimate surface fluxes by optimising prior emission inventories to match observed atmospheric CH₄ concentrations (Patra et al., 2018). Recent studies have increasingly turned to data-driven machine-learning models. For instance, (McNicol et al., 2023) upscaled eddy covariance
- 140 CH₄ fluxes globally using random forest algorithms, while (Bernard et al., 2025) introduced a satellite observation-based model to simulate temporal emission variability. These machine-learning approaches offer distinct advantages, including rapid operational speeds and lower computational costs. However, they are constrained by the sparsity of training data.
- 145 Building on these well-established controls and the advancements in data-driven modeling, we provide a low-latency continuation of the GMB gridded wetland CH₄ emissions by extending monthly emissions from 2000-2020 through 2025 at 1°x1° resolution. This approach is supported by the fact that many key climate, hydrometeorological and vegetation drivers are routinely updated as global gridded datasets, enabling a practical emulator-based pathway for timely wetland CH₄ emission estimates. In
- 150 this study, wetlands are defined following the GMB natural wetland category, excluding lakes, rivers, reservoirs, coastal waters and managed sources (Saunois et al., 2025). We emulate each BU and TD ensemble member from GMB using machine-learning models trained on the 2000-2020 GMB wetland emission fields and climate predictor variables (e.g., soil temperature and precipitation) from ERA5 reanalysis to yield spatially explicit monthly emissions for 2000-2025 with uncertainty derived from
- 155 both GMB ensemble spread (35 runs) and ERA5 monthly reanalysis (10 ensemble members). Using the resulting 2000-2025 wetland CH₄ emission data, we then quantify recent emission changes in 2021-2025 relative to 2000-2020, characterize interannual variability in annual regional and latitudinal emission time series, and diagnose long-term growth rates and their seasonality.
- 160 This proposed framework and dataset provide a highly responsive and scalable tool for contemporaneous, spatially explicit wetland CH₄ emission estimates. It provides support for early-warning diagnostics, attribution of recent atmospheric CH₄ anomalies, and provision of timely priors for atmospheric inversions.



165 2 Methods

2.1 Input datasets

2.1.1 Wetland CH₄ flux data

We used wetland monthly net CH₄ flux estimates from the most recent Global Methane Budget (GMB) synthesis for 2000-2020, including both process-based models (“bottom-up”, BU) and atmospheric
 170 inversions (“top-down”, TD) (Saunois et al., 2025; Zhang et al., 2025). Detailed descriptions of each model and inversion framework are provided in the cited GMB publications; here we summarize the key elements relevant to this study.

The BU wetland biogeochemical models (22 estimates, Table S1) included in the analysis are from 11
 175 models: CLASSIC (Arora et al., 2018; Melton and Arora, 2016), ELM-ECA (Riley et al., 2011), ISAM (Shu et al., 2020; Xu et al., 2021), JSBACH (Kleinen et al., 2020, 2021, 2023), JULES (Gedney et al., 2019b), LPJ-MPI (Kleinen et al., 2012), LPJ-WSL (Zhang et al., 2016), LPX-Bern (Spahni et al., 2011; Stocker et al., 2014), ORCHIDEE (Ringeval et al., 2011), SDGVM (Beerling and Woodward, 2001; Hopcroft et al., 2011, 2020), and VISIT (Ito and Inatomi, 2012). Each model was run under two global
 180 climate forcings (CRU and GSWP3-W5E5). All BU simulations are prognostic, i.e. each model computes wetland extent internally rather than relying on a shared wetland extent dataset. Differences in simulated wetland extent therefore contribute to inter-model spread in CH₄ emissions and may explain a substantial fraction of discrepancies across regions.

TD inversion products (13 estimates, Table S2) were constrained by surface or satellite observations
 185 over the 2000-2020 period and they are from seven systems: CarbonTracker Europe-CH₄ (Tsuruta et al., 2017), LMDz-CIF (Thanwerdas et al., 2022), LMDz-PYVAR (Zheng et al., 2018a, b, 2019), MIROC4-ACTM (Chandra et al., 2021; Patra et al., 2018), NISMOM-CH₄ (Niwa et al., 2022, 2025), and NIES-TM-FLEXPART (NFVAR) (Maksyutov et al., 2021; Wang et al., 2019). Each inversion system was
 190 driven by two anthropogenic emission inventories (two separate estimates): EDGAR v6 (Crippa et al., 2021) and GAINS (Höglund-Isaksson et al., 2020), except NIES-TM-FLEXPART (GOSAT) which was run only under EDGAR v6. Wetland priors were prescribed as dynamic emissions from the ensemble mean of 11 wetland models used in the previous GMB estimates (Saunois et al., 2020). We used
 195 posterior flux estimates throughout this study.

All BU and TD outputs provide monthly wetland CH₄ flux as emission per grid cell area, typically at
 1°x1° resolution in the regridded GMB output products. Coarser products (CLASSIC models at 1.85°
 and JSBACH models at 2.8°) were remapped to a 1°x1° grid for spatial consistency across datasets. In
 total, we included 22 BU estimates and 13 TD estimates (35 estimates overall). Tables S1 and S2
 200 summarize the BU and TD estimates.



2.1.2 Predictor variables from ERA5 reanalysis

We assembled 10 ensemble members of global ERA5 monthly averaged data at $0.5^\circ \times 0.5^\circ$ spatial resolution (Hersbach et al., 2023) for 11 climate predictor variables: t2m (near-surface air temperature), tp (total precipitation), ssrd (downward shortwave radiation at surface), slhf (surface latent heat flux), sshf (surface sensible heat flux), swvl1 (volumetric soil water in layers 1, 0-7 cm below surface), swvl2 (volumetric soil water in layers 2, 7-28 cm below surface), stl1 (soil temperature in layer 1, 0-7 cm below surface), stl2 (soil temperature in layer 2, 7-28 cm below surface), lai_hv (high-vegetation leaf area index), and lai_lv (low-vegetation leaf area index). LAI from ERA5 is prescribed in the ECMWF land-surface model as a seasonally varying, monthly climatology derived from satellite LAI (Roberts et al., 2018). We chose ERA5 data in part because it is operationally updated regularly, which enables continuous extension of predictor fields and low-latency updating of emission reconstructions as new months become available. ERA5 data has also undergone extensive evaluation and is widely used due to its high spatiotemporal consistency and generally strong performance (Hersbach et al., 2020), but notably regional biases remain, for example, a weaker performance in parts of Asia/Africa for hydrology (Gebrechorkos et al., 2024) and temperature in the Arctic (Tian et al., 2024). The 10 ensemble members correspond to the ERA5 Ensemble of Data Assimilations, which provides multiple physically consistent realizations of the reanalysis by perturbing the data assimilation system to sample uncertainty in atmospheric and land-surface states. The ensemble spread can be interpreted as an indicator of reanalysis uncertainty.

The 11 variables included in this analysis represent first-order climatic and biophysical controls on wetland CH₄ production, oxidation, and transport and their relevance to wetland CH₄ emissions is well established in the literature (Knox et al., 2021; Li et al., 2024, 2025; McNicol et al., 2023; Pu et al., 2024; Toet et al., 2011; Yuan et al., 2022, 2024; Zhang et al., 2025). To represent the ecosystem memory effect in response to environmental conditions (Chen et al., 2025), we also included 1-month and 2-month lagged versions of each variable. The data are regridded using bilinear interpolation to $1^\circ \times 1^\circ$ resolution to match the wetland CH₄ flux data from the regridded GMB models for further machine-learning modeling.

We recognize that many of these variables are multicollinear, an issue that XGBoost is considered robust against. Nevertheless, we minimized overfitting by constraining model complexity (feature screening, regularized trees with subsampling), using a held-out validation window with early stopping to select the optimal number of boosting iterations, and evaluating performance on temporally separated test periods (2000-2002, 2018-2020) not seen during training (see section 2.2.2 Model Architecture). In addition, using the 10 ERA5 ensemble members allows us to propagate reanalysis uncertainty in the predictor fields through the reconstruction.



2.2 XGBoost model

2.2.1 Data assembly and filtering

For each of 350 model-ensemble pairs (35 GMB model estimates x 10 ERA5 ensemble members), we constructed gridded monthly time series pairing GMB data with a set of 35 predictor variables: 11 contemporaneous ERA5 variables, 22 lagged variables (1- and 2-month lags for each of the 11 ERA5 variables), and two seasonal encoding terms:

$$\text{mo_sin} = \sin(2\pi \text{ mo}/12), \text{ mo_cos} = \cos(2\pi \text{ mo}/12) \quad (\text{Eq.1})$$

Where:

mo=month of year (1-12, 1=January and 12=December);

π =the constant pi;

mo_sin and mo_cos are the sine/cosine transforms of mo and represent seasonality.

We optimised our analyses to reduce noise and computational burden by retaining only grid cells with substantial wetland emissions, defined as those maintaining a mean flux of 1×10^{-15} kg CH₄/m²/s or greater across all GMB model estimates over 2000-2020 (30,000 times lower than the global mean).

2.2.2 Model architecture

To systematically reconstruct wetland CH₄ fluxes, we developed an machine-learning pipeline, training independent Extreme Gradient Boosting (XGBoost) models for each grid cell and model-ensemble pair. XGBoost has been applied as a surrogate model for ensemble estimates of GMB BU models with high performance (Zhu et al., 2024). The overall modeling workflow consisted of four stages: (1) data splitting, (2) feature screening, (3) hyperparameter running, and (4) final model fitting and prediction. Each stage is introduced as follows.

Each model-ensemble pair dataset was temporally split into a training set and two testing sets to assess reconstruction capability across different climatic periods at monthly scale. The training data cover January 2003 to December 2017. The testing data cover two periods: January 2000-December 2002 and January 2018-December 2020. This splitting strategy is to test the model's ability to capture long-term climate variabilities in early/late periods and reduce overfitting. We implemented an early stopping mechanism using a hold-out validation approach. The last 24 months of the training period were reserved as a validation set (January 2016-December 2017).

Prior to XGBoost training, we applied Boruta feature screening (Kursa and Rudnicki, 2010) on the training subset to reduce the effective dimensionality of the predictor set and mitigate overfitting risk arising from the relatively large number of candidate predictors. Boruta augments the predictor matrix with 'shadow' features formed by randomly permuting each predictor across samples, thereby preserving each predictor's marginal distribution while separating its association with CH₄ flux. A random forest regressor is fit to the combined real/shadow feature set and feature importances are



compared. Predictor variables were retained if their random forest importance exceeded the maximum shadow-feature importance in more than 20% of 10 iterations (Boruta-style screening), while seasonal terms were always retained. The resulting per-grid cell feature subset was then used for both XGBoost fitting and subsequent reconstructions, ensuring that training and prediction used an identical, cell-specific predictor variable set.

After the feature screening, we performed a grid search over six hyperparameter configurations (learning rate, maximum tree depth, minimum child weight, subsampling ratio, column subsampling per tree, and L2 regularization) (Table S3). For each configuration, training stopped if the validation Root Mean Squared Error (RMSE) did not improve for 50 consecutive boosting rounds. Once the optimal hyperparameter set was identified for a grid cell, the model was refitted on the full training data using the optimal number of boosting iterations selected during early stopping.

Once the optimal hyperparameter combination was identified for a specific grid cell, the model was refitted on the training data using the optimal number of boosting iterations determined during the search. The saved booster was then applied to predict wetland CH₄ flux 2000-2025 (including 2021-2025, which extends beyond the most recent GMB estimates period). All the predicted wetland fluxes (unit of kg CH₄/m²/s) were converted to emissions (Tg CH₄/month) by multiplying the fluxes by grid-cell area (m²) and total number of seconds (s) in each month.

To summarize the mitigation measures for potential overfitting: we (i) use a strict temporal split with training (2003-2017), validation (last 24 months of training) for early stopping, and two disjoint withheld test windows (2000-2002 and 2018-2020), and (ii) constrain model complexity via regularization and subsampling.

2.2.3 Model evaluation

Model performance was evaluated for each grid cell per model-ensemble pair by computing the R² and RMSE values on the withheld test periods. The mean R² and RMSE of pairs per grid cell are applied to evaluate model performance at the grid cell level (Figure 1). To assess the model reconstruction skills for grid cells with different emission magnitudes, we grouped grid cells into (1) emission percentile bins based on their total wetland CH₄ emissions during 2000-2020 from GMB estimates, (2) 18 geographical regions, and (3) five latitude bands. We then compared the mean R² and RMSE values of each emission bin (0-10% denotes the highest emitting grid cells) and 18 geographical regions (Figure S1, S2). Monthly mean R² and RMSE at five latitude bands over test periods are presented to study the temporal variation of model performance (Figure S3, S4). Detrended monthly global emissions from GMB and predictions were compared to assess the model performance in capturing interannual variability (Figure S5). We generated out-of-sample wetland CH₄ emission predictions (CH₄_pred) for the test periods to compare with the emissions from GMB models (CH₄_GMB) (Figure 2). This ensured that the model predicted CH₄_pred based on the trained model without having seen the CH₄_GMB in the test data.



2.3 Emission trend analyses

We quantified spatial patterns, regional aggregates and temporal emission trends in reconstructed wetland CH₄ emissions for 2000-2025, and compared with corresponding GMB estimates over 2000-2020. We performed analyses for 18 geographic regions defined in Saunois et al., 2025 and for five latitude bands. The 18 regions are USA, Canada, Central America, Northern South America, Brazil, Southwest South America, Europe, Northern Africa, Equatorial Africa, Southern Africa, Russia, Central Asia, Middle East, China, Korea-Japan, South Asia, Southeast Asia, and Australasia. Five latitudinal regions are the global, Northern Hemisphere (NH) high latitudes (60°N-90°N), NH mid-latitudes (30°N-60°N), tropics (30°N-30°S), and Southern Hemisphere (SH) extratropics (30°S-90°S).

We analyzed recent emission changes by comparing 2021-2025 against 2000-2020 baseline at both grid-cell and regional scales (Figures 3, 4). For Figure 3, gridded monthly emissions were aggregated to per-grid mean emissions for each period (2021-2025 and 2000-2020). Emissions presented in Figure 4 were summed across grid cells within each region. We examined interannual variability and long-term annual emission changes by aggregating monthly emissions to annual totals for both the latitude bands and 18 geographic regions (Figures 5, S3, S5). Ordinary least squares regression was applied to monthly emissions over 2000-2025 to estimate emission trends (Figure 6).

3 Results and Discussion

3.1 Model performance evaluation

Across the two test windows (2000-2002 and 2018-2020), the XGBoost model reproduces monthly wetland CH₄ emissions with high skill. Globally, the coefficient of determination R^2 is 0.65 ± 0.003 (mean $\pm$ 95% CI, hereafter; Figure 1a), indicating strong predictability from predictor variables. However, the model performance varies geographically. Mean R^2 is higher in the Southern Hemisphere (SH) than in the Northern Hemisphere (NH) (0.77 ± 0.006 vs 0.62 ± 0.004), and higher in the tropics than in the extratropics (0.78 ± 0.004 and 0.58 ± 0.004 , respectively). Mean R^2 is higher in regions with higher wetland CH₄ emissions (“top emitting regions”) (Figure S1a). Global RMSE over test periods is $5.49 \pm 0.12 \times 10^{-3}$ Tg CH₄/year (Figure 1b). RMSE for the tropics is about twice of the RMSE for the extratropics ($9.37 \pm 0.27 \times 10^{-3}$ and $3.35 \pm 0.08 \times 10^{-3}$ Tg CH₄/year, respectively). Top emitting regions exhibit higher RMSE, e.g., RMSE doubled for top 10% emitting regions compared with top 10-20% emitting regions (Figure S1b). At geographical scale, mean R^2 is highest in Southeast Asia and Korea-Japan (>0.8), and lowest in Russia and Canada (~ 0.5) (Figure S2a). Mean RMSE is highest in America (Northern South America and Central America), and lowest in the Middle East and Central Asia. Temporal variation of model performance reveals the lowest R^2 in Jan/Feb in NH high latitudes (Figure S3), and the highest RMSE in July-Sep in the tropics (Figure S4). We further evaluate how these spatial and temporal differences in model skill influence long-term variability in annual emissions in Section 3.3.

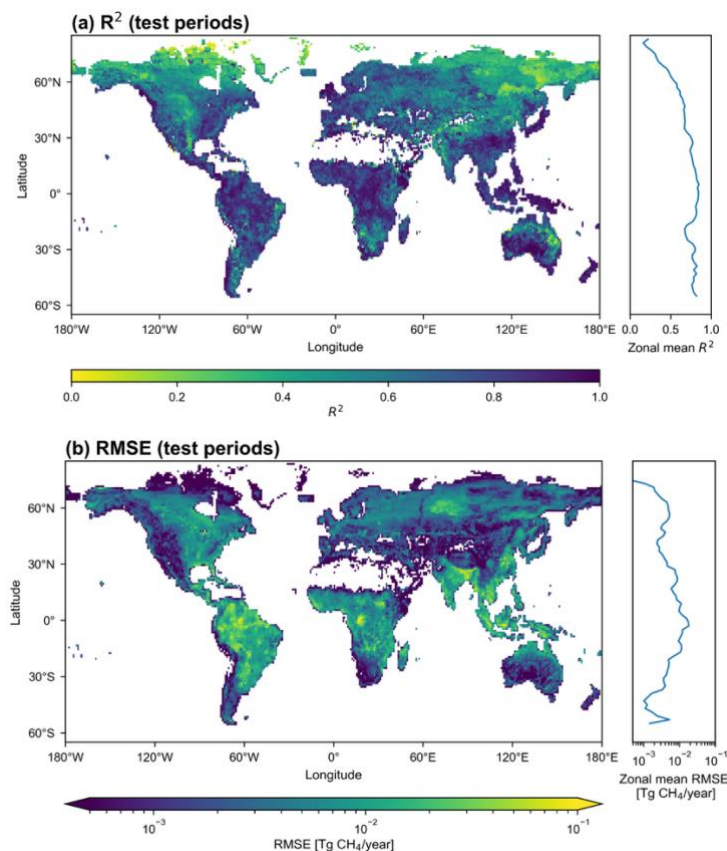


Figure 1. Predictive skill of the XGBoost wetland CH_4 model (R^2 and RMSE for test periods). Insets indicate the zonal mean R^2 and RMSE in 2° latitude bands.

To examine predictive uncertainty, we compare wetland CH_4 emissions from the GMB models (CH_4 _GMB; Figure 2a) with XGBoost predictions (CH_4 _pred; Figure 2b) over the two test periods. Globally, CH_4 _pred reproduces both the magnitude and spatial pattern of CH_4 _GMB, including the major hotspots concentrated in the tropics, such as Amazon Basin (central-western Amazon floodplains, Pantanal and Llanos de Moxos), Congo Basin (Cuvette Centrale and the Sudd of South Sudan), and Southeast Asia (Sumatra, Borneo, Peninsular Malaysia, and New Guinea), with additional hotspots in the northern high latitudes, such as West Siberian Plain and Hudson/James Bay Lowlands.

The differences in emissions (Figure 2c) and the corresponding percent emission differences (Figure 2d) between CH_4 _GMB and CH_4 _pred highlight spatial biases. The global mean predicted wetland CH_4 emission is 2.27 Tg/year lower than the global mean of CH_4 _GMB during test periods (Figure 2c), corresponding to a 1.41% underestimation. Underestimation is strongest over high-emission systems, such as the central-western Amazon, Southeast Asia, Congo Basin and West Siberian Plain (Figure 2c), where absolute emission differences are negative, but percent differences are smaller because the



denominator (CH_4_GMB) is large. For example, in Southeast Asia, the mean bias is -0.44 Tg/year , yet the relative bias is only -1.74% given a high regional mean emission (CH_4_GMB of 25.23 Tg/year). Monthly anomalies in global emissions reveal small disagreements between the predictions and GMB estimates ($\sim \pm 4 \text{ Tg CH}_4/\text{year}$) (Figure S5). Largest discrepancies occur in July/August 2020 (underestimation of $15 \text{ Tg CH}_4/\text{year}$).

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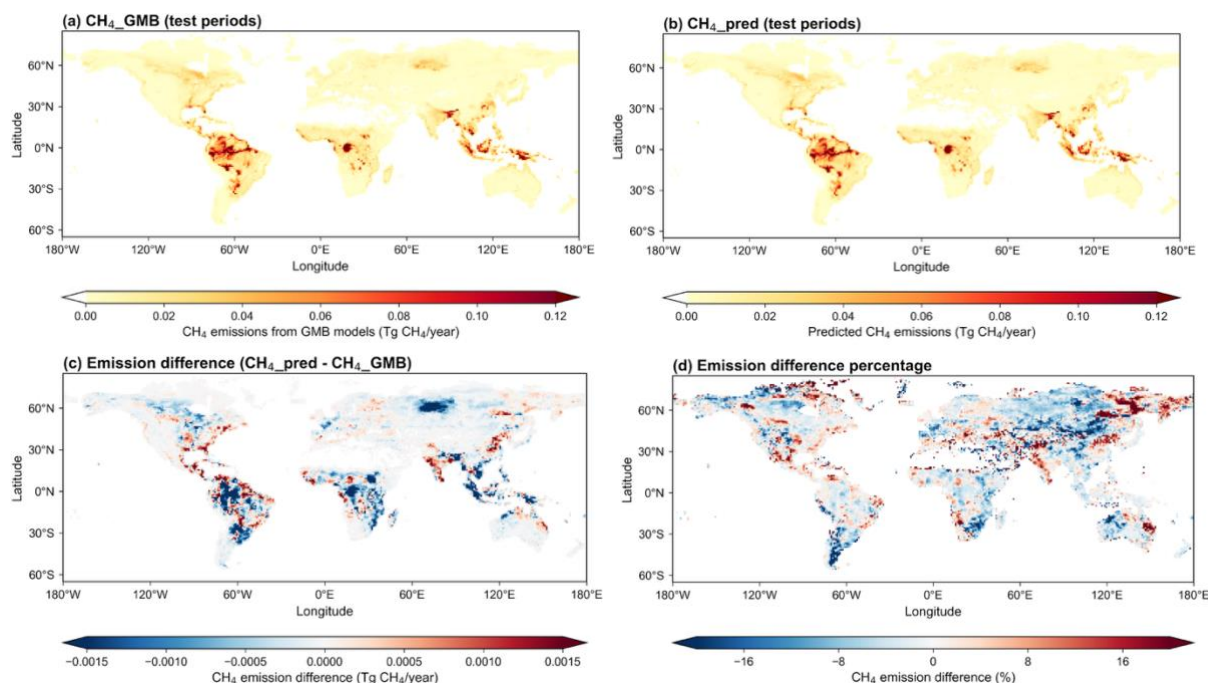


Figure 2. Wetland CH_4 emissions (per $1^\circ \times 1^\circ$ grid cell) from GMB models and XGBoost predictions for two test periods combined. (a) Mean CH_4_GMB (GMB emissions) over all test months. (b) Mean CH_4_pred (XGBoost prediction) over all test months. (c) Differences in mean CH_4_GMB and mean CH_4_pred . (d) Percent emission difference between mean CH_4_GMB and mean CH_4_pred . Reds in (c) and (d) indicate higher emissions in CH_4_pred , blues lower emissions.

390 3.2 Emission trends and predicted anomalies in recent years (2021-2025)

Figure 3 shows spatial maps of global predicted mean wetland CH_4 emissions for 2021-2025 (Figure 3a) and for 2000-2020 (Figure 3b). The 2021-2025 mean emission is $157.83 \pm 2.38 \text{ Tg/year}$ (z-score: 3.72), representing a weak increase of 0.05 Tg/year relative to the 2000-2020 mean. However, changes are more pronounced across latitudes. NH mid- and high-latitude wetland emissions increase during 2021-2025 by 0.76 ± 0.07 (from 29.92 ± 1.11 (2000-2020) to 30.68 ± 1.15 (2021-2025)) (z-score: 2.21) and 0.35 ± 0.03 (from 11.04 ± 0.68 to 11.39 ± 0.70) Tg/year (z-score: 1.01), respectively, compared with



2000-2020 emissions. These changes correspond to $\sim 2.5\%$ and $\sim 0.8\%$ increase. The continued growth in these regions in 2021-2025 (relative to 2000-2020, this study), together with increases reported for 2010-2018 relative to 1981-1989 (Feron et al., 2024), suggests an increasing role of boreal and temperate wetlands in recent decades of global warming. In contrast, emissions decrease in the tropics and SH extratropics during 2021-2025. The tropics exhibit a 0.95 ± 0.19 Tg/year emission decrease (z-score: -2.81), from 113.89 ± 2.46 (2000-2020) to 112.94 ± 2.49 Tg/year (2021-2025), while SH extratropics emissions decline by 3.5% (-0.11 ± 0.02 Tg/year, z-score: -0.34) from 3.06 ± 0.12 (2000-2020) to 2.96 ± 0.12 Tg/year (2021-2025).

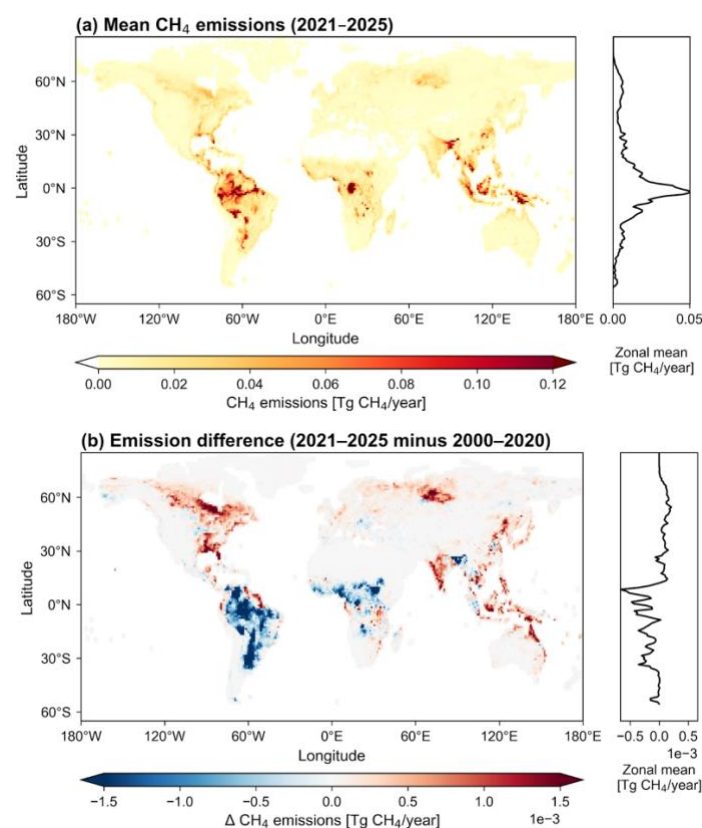


Figure 3. (a) Mean predicted wetland CH₄ emissions for 2021-2025. (b) Difference between mean emissions in 2021-2025 and 2000-2020. Reds indicate higher emissions in 2021-2025, blues lower. Insets indicate the zonal mean in 2° latitude bands.

We further assess regional wetland CH₄ emissions for 2021-2025 from XGBoost predictions (PRED 21-25, dark blue bars in Figure 4) across 18 regions, and compared them with (i) CH₄ emissions for 2000-2020 from GMB models (GMB 00-20, light green bars) and (ii) XGBoost model predictions for 2000-2020 (PRED 00-20, dark green bars). Four regions show no significant change (labelled “n.s.” above



PRED 21-25) when comparing 2021-2025 with 2000-2020 using XGBoost predictions. Among the 14 regions with significant changes, nine regions show increases and five show decreases (Northern South America, Brazil, Southwest South America, Northern Africa, and Equatorial Africa). Notably, all regions with decreases are located in South America and Africa. The largest emission changes (2021-2025 relative to 2000-2020) occur in Brazil (-0.59 ± 0.09 Tg/year), Canada ($+0.46 \pm 0.05$ Tg/year), Southwest South America (-0.43 ± 0.07 Tg/year), Russia ($+0.31 \pm 0.03$ Tg/year), Southeast Asia ($+0.22 \pm 0.03$ Tg/year) and Equatorial Africa (-0.21 ± 0.04 Tg/year). The pronounced decline over Brazil possibly reflects the exceptional drought conditions in the Amazon basin during 2022-2024, including record-low river levels and anomalously warm, dry conditions (Espinoza et al., 2024). Such hydroclimatic drying reduces floodplain inundation and lowers water tables, which suppresses anaerobic conditions and thereby limits wetland methane production and emissions (Cui et al., 2024).

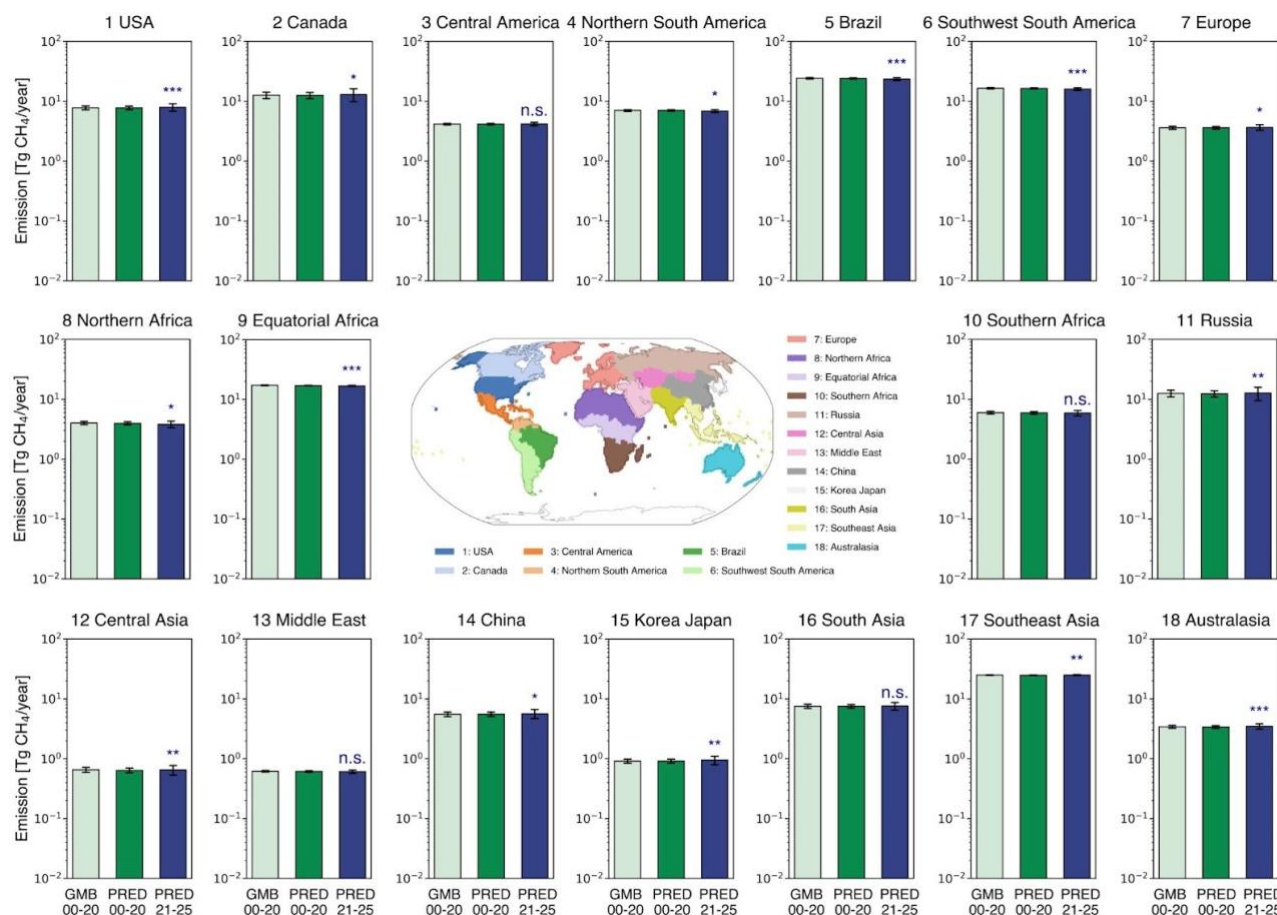


Figure 4. Regional wetland CH₄ emissions for 2000-2020 and 2021-2025 across 18 regions. Bars show regional mean emissions: GMB 00-20 (GMB estimates, 2000-2020; light green), PRED 00-20 (XGBoost predictions, 2000-2020; dark green), PRED 21-25 (XGBoost predictions, 2021-2025; dark blue). Error bars show 95% CI. A two-sided paired test was applied for each region to assess whether the PRED21-25 mean differs from the PRED



435 00-20 mean. Regions with $p \geq 0.05$ are labelled “n.s.” above the PRED 21-25 bar. Asterisks above PRED 21-25 bars denote significance: *** for $p < 0.001$, ** for $p < 0.01$, * for $p < 0.05$.

3.3 26-year emission trends and anomalies (2000-2025)

440 The emission predictions enable an assessment of long-term trends and interannual variability in wetland emissions over 2000-2025. Annual emission time series show pronounced peaks in 2011 and 2016/2017 in the tropics, NH mid-latitudes, and globally (Figure S6). Monthly emission anomalies in these regions (Figure S7) indicate that the annual maxima are driven by sustained positive anomalies during Aug 2010-May 2011 and Aug 2016-Dec 2017, which coincide with La Niña conditions. This
 445 alignment is consistent with prior studies reporting La Niña-driven wetting and expanded inundation that enhance tropical wetland CH₄ emissions (Hodson et al., 2011; Lin et al., 2024; Murguía-Flores et al., 2023; Zhang et al., 2020, 2018; Zhu et al., 2017).

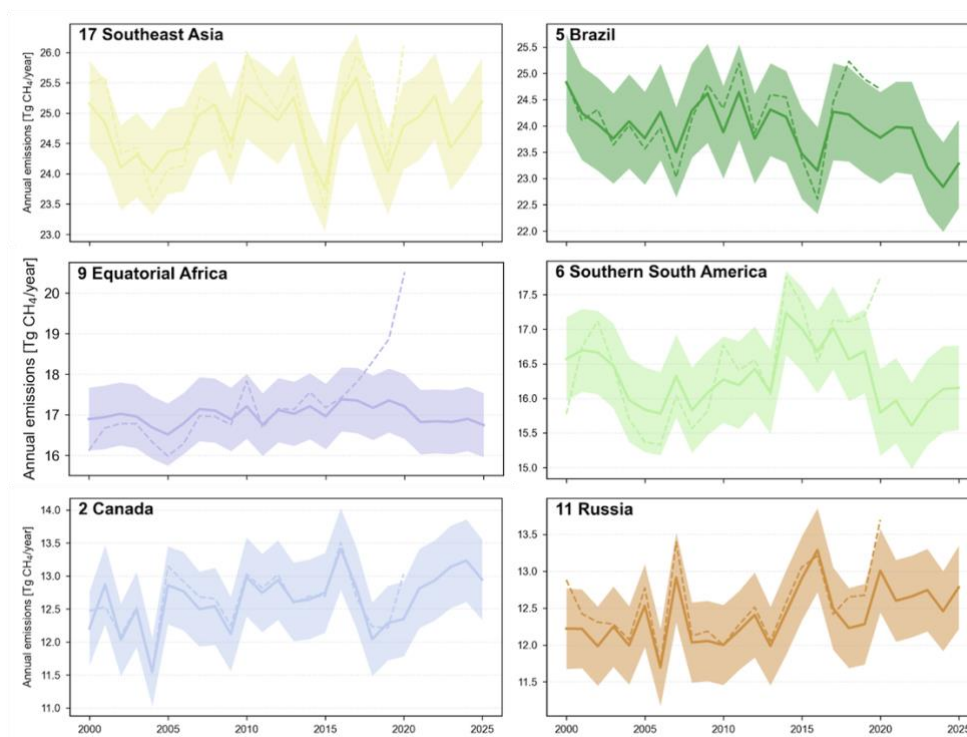
In the 2020s, global and tropical annual emissions reach a minimum in 2023 (Figures S3, S4),
 450 coincident with a strong El Niño event. This is supported by two recent studies reporting a sharp decline in wetland emissions in South America (Ciais et al., 2026) and Amazonia (Quinn et al., 2025) in 2023 linked to El Niño-related drought. Our results further suggest that Brazil and Southern South America are the key contributors, and both rank among the top six emitting regions globally (Figure 5). Annual emissions in Brazil decline steadily across the study period, from 24.83 ± 0.93 Tg/year in 2000 to 23.28 ± 0.84 Tg/year in 2025, with the lowest emission in 2024 (22.84 ± 0.85 Tg/year) (Figure 5). Emissions
 455 in Southern South America are relatively stable during 2000-2014, decrease markedly during 2014-2023 (from 17.24 ± 0.61 to 15.95 ± 0.62 Tg/year), and show a modest recovery in 2024/2025 (Figure 5).

The remaining four top-emitting regions show contrasting behavior. Since 2019, Southeast Asia,
 460 Canada and Russia show increasing emissions, whereas Equatorial Africa emissions decrease slightly (Figure 5). Southeast Asian emissions fluctuate around 24.74 ± 0.48 Tg/year over 2000-2025, with a minimum in 2015 (23.75 ± 0.70 Tg/year) and a maximum in 2017 (25.58 ± 0.73 Tg/year). Emissions in Canada are comparatively stable during 2005-2015, while variability is larger before 2005 and after 2015: the amplitude reaches ~ 1.5 Tg/year in these periods, approximately double that during 2005-2015 (~ 0.8 Tg/year). Equatorial African emissions remain 17.0 ± 0.2 Tg/year during 2000-2025, despite two
 465 notable declines during 2002-2005 (-0.51 Tg/year) and 2019-2025 (-0.61 Tg/year).

Regional emission time series (Figures 5, S8) help explain the pronounced disagreement in 2019-2020 for global and tropical emissions, where predictions are substantially lower than the GMB estimates
 470 (Figure S6). For 2019, prediction mean are 5.53 Tg/year below the GMB estimates for global emissions ($\sim 3.5\%$ underestimation), while emission changes in Equatorial Africa, Brazil, USA, Southern Africa, and Southern South America jointly account for 4.26 Tg/year of this discrepancy. In 2020, the modeled mean global emissions are 9.84 Tg/year lower than GMB ($\sim 6\%$ underestimation). The discrepancies in the tropics explain 9.17 Tg/year of the underestimation, including major regions such as Eastern Africa,



475 Southern South America, Southeast Asia, Brazil, and Northern Africa. Notably, this disagreement in
 2019-2020 is not evident in other latitude bands and does not persist outside these years.



480 Figure 5. Annual wetland CH₄ emissions for the six top-emitting regions. Solid lines show model predictions of
 annual mean emissions, and shaded areas indicate 95% CI. Dashed lines show GMB emission estimates (mean).
 Panel titles indicate the region code and region name, which correspond to the region definitions shown in Figure
 4.

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Finally, we examine trends in predicted monthly wetland CH₄ emissions across five latitude bands
 (Figure 6a) and 12 calendar-month categories (Figure 6b) over 2000-2025. We summed predicted
 monthly emissions across all grid cells within each latitude bands to form regional monthly time series,
 490 then computed monthly emission anomalies by subtracting the 2000-2020 climatological mean. We
 then estimated growth rates using ordinary least squares linear regression of the anomaly time series.
 Growth rates presented in Figure 6 indicate the total emission changes over 26 years.

The global wetland CH₄ emission growth rate is 2.15 ± 0.89 Tg/year (26 years total, z-score: 4.73)
 495 (Figure 6a). Positive trends are concentrated in the NH, with the largest increase in the mid-latitudes
 (1.44 ± 0.37 Tg/year, z-score: 7.73) followed by the high-latitudes (0.81 ± 0.24 Tg/year, z-score: 6.70).
 In contrast, SH extratropics exhibit weak negative growth rates (-0.10 ± 0.09 Tg/year, z-score: -2.21).

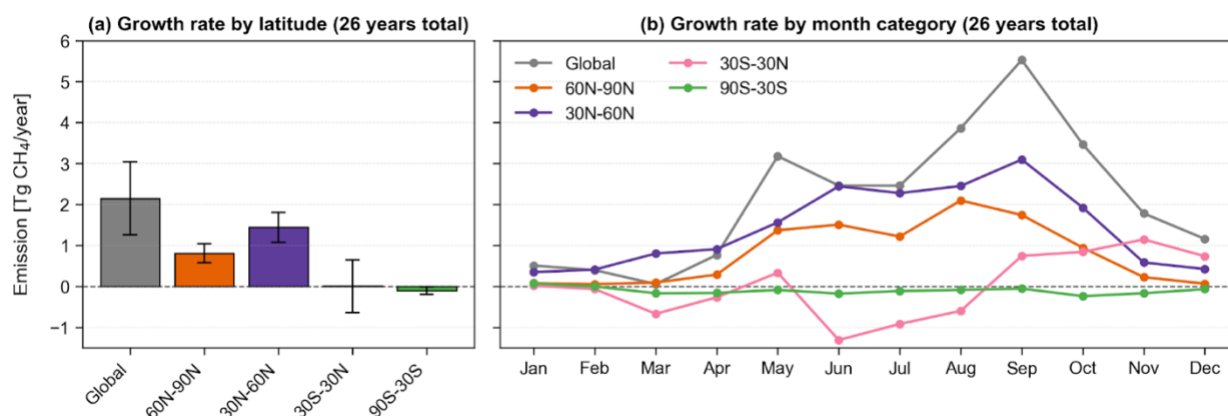


The growth rate for the tropics is close to zero with higher uncertainty ($5.05 \times 10^{-3} \pm 0.64$ Tg/year, z-score: 0.02).

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Evaluating growth rates over individual months (Figure 6b), we find that the global growth rate peaks in late boreal summer, with the largest increases in August and September (2.45 ± 2.24 and 3.10 ± 1.44 Tg/year, respectively). The strongest seasonal intensification occurs in NH mid-latitudes during June-September ($2.27 - 3.10$ Tg/year), consistent with the dominant contribution from boreal growing-season emissions. For both NH mid- and high-latitudes, growth rates are systematically higher during May-October (growing season) than during November-April. In the tropics, monthly growth reaches its highest in Oct-Dec and lowest in Jun-Aug (negative growth). SH extratropics show near-zero growth throughout the year (slightly negative).

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Figure 6. Regional and seasonal growth rates (Tg/year) in predicted wetland CH₄ emissions for 2000-2025. (a) Growth rates across five latitude bands over the full 26-year period. (b) Growth rates for each calendar month across the same period. Positive growth rates indicate an increase in CH₄ emissions, while negative values indicate a decrease. Error bars indicate 95% CI.

515

4 Conclusions

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This study presents a framework to extend global natural wetland CH₄ emissions from 2000 through 2025 by applying a machine-learning emulator to disentangle the release of GMB estimates to estimate wetland emissions at low latency, with an operational design intended for routine updates. The extended wetland CH₄ emission record shows that the magnitude of emission changes in 2021-2025 relative to 2000-2020 is substantially larger than the long-term trend over 2000-2025, even though the global mean emission change is relatively small.

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A key emerging feature is accelerated emissions in the NH extratropics, with the strongest increases in mid-latitudes and a clear positive signal in high latitudes. Seasonal trend analyses indicate that these



increases are concentrated in the warm season, consistent with an amplification of the seasonal cycle and a strengthening of late-summer growth, highlighting Northern mid- and high-latitude wetlands as regions where intensifying trends are most evident in recent years.

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In contrast, the tropics and SH extratropics show weakly negative emission changes in post-2020. Nevertheless, the tropics still dominate interannual variability of global wetland emissions and can strongly influence the global CH₄ budget through episodic hydroclimate extremes.

535 Overall, this work proposes an operational, low-latency emulator framework that provides a scalable pathway to track wetland CH₄ emissions in response to climate anomalies by leveraging routinely updated Earth observation reanalysis data.

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Data Availability

ERA5 monthly averaged data are available at Copernicus Climate Change Service Climate Data Store at <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels-monthly-means?tab=overview> (DOI:10.24381/cds.f17050d7) (Accessed on 22-Jan-2026) (Copernicus Climate Change Service, 2023).

545 The global natural wetland methane emission dataset generated from this study are publicly available at <https://doi.org/10.5281/zenodo.18870108> (Li et al., 2026).

Competing Interests

At least one of the (co-)authors is a member of the editorial board of Earth System Science Data.

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